



Assembly Principles and Quality metrics: Reference vs De novo

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Key terminologies

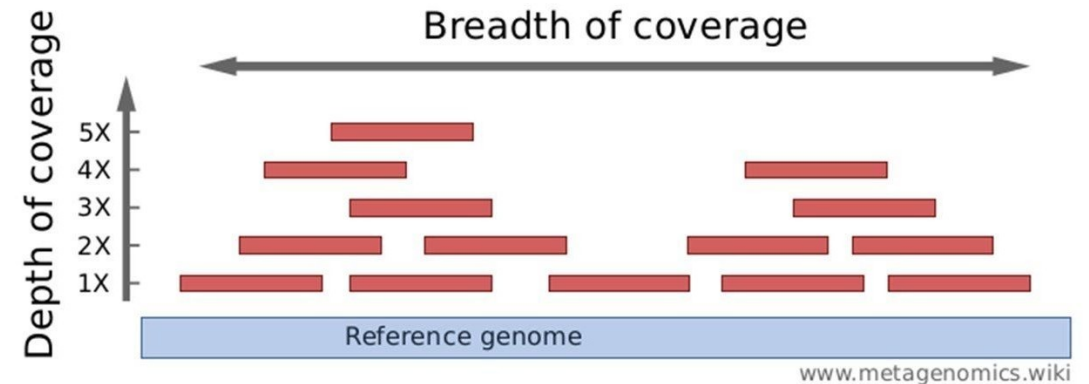
Base-pair - basic building block of double-stranded DNA, unit of DNA segment length (bp)

Read - continuous sequence produced by sequencer

Read depth - the number of short reads that overlap each other within a specific genomic region (how many times the particular base or region is read)

Genome Coverage - the proportion of the entire genome that has been sequenced and adequately represented in a DNA sequencing experiment

Sequencing Depth vs Coverage



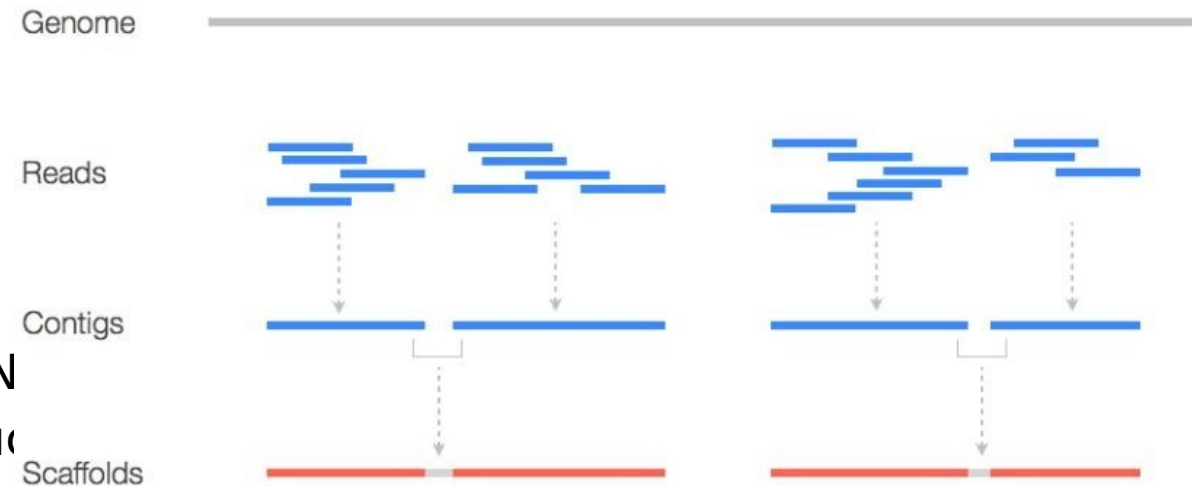
Key terminologies

Consensus sequence - representation of a DNA or RNA sequence that incorporates the most common or prevalent base at each position in a set of related sequences

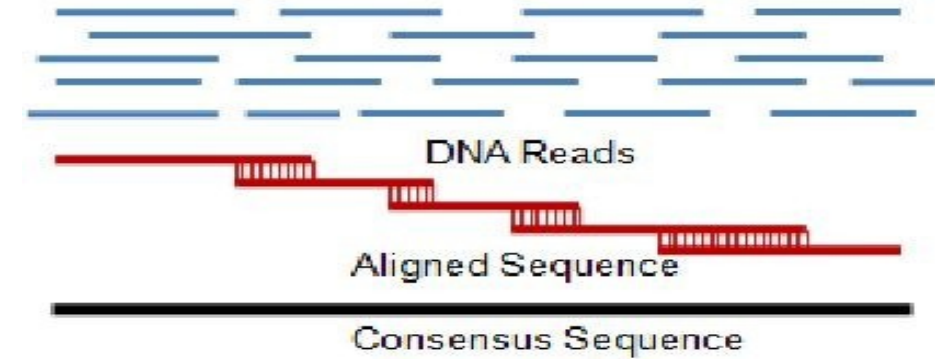
Contig - set of overlapping segments (reads) of DNA sequences forming continuous consensus sequence

Assembly - aligning and merging fragments of DN sequence (reads, contigs) in order to reconstruct the original sequence

Scaffold - set of linked non-contiguous series of genomic sequences, consisting of contigs separated by gaps of roughly known length



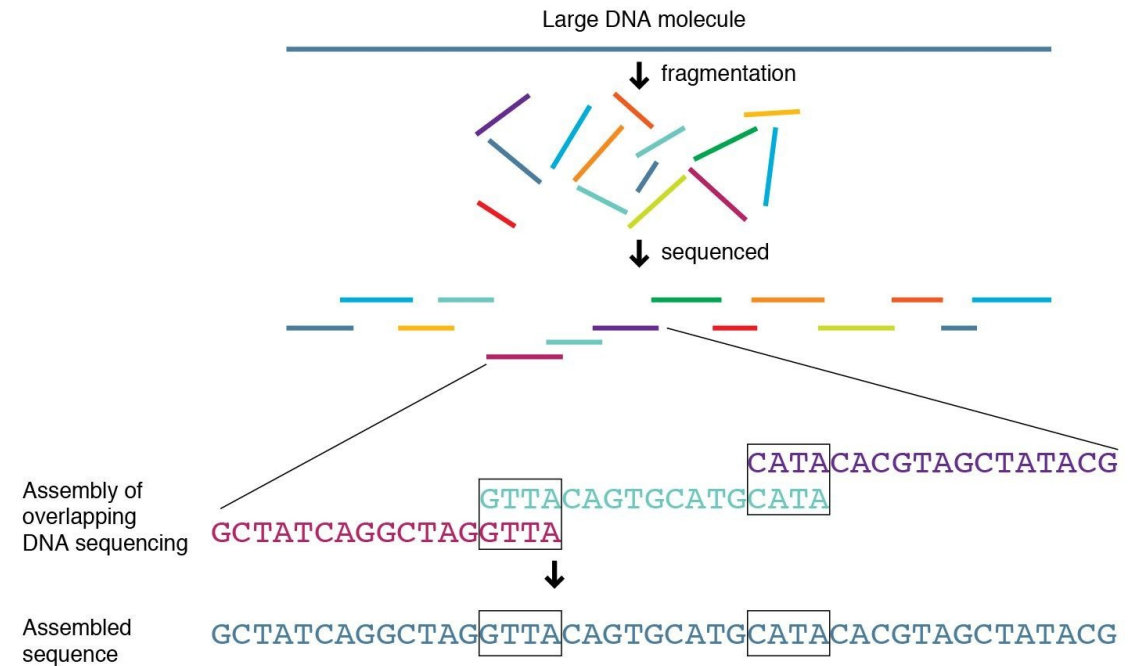
Genome Reconstruction



- Many areas of research in computational biology rely on the availability of complete whole-genome sequence data.
- Yet the process to sequence a whole genome is itself non-trivial and an area of active research.
- The problem lies in the fact that current genome-sequencing technologies cannot continuously read from one end of a long genome sequence to the other.
- They can only accurately sequence small sections of base pairs (ranging from 100 to a few thousand, depending on the method), called reads.
- Therefore, in order to construct a sequence of millions or billions of base pairs (such as the human genome), computational biologists must find ways to combine smaller reads into larger, continuous DNA sequences

Assembly

- There are two approaches for genome assembly: **reference-based** (or comparative) or **de novo**.
- In a reference-based assembly, we use a reference genome as a guide to map our sequence data to and thus reassemble our sequence this way.
- We can as well create a 'new' (**de novo**) assembly that does not rely on a map or reference and more closely reflects the actual genome structure of the isolate that was sequenced.



Pairwise Sequence alignment

Global alignment - Matching as many characters as possible across their entire length

- The tool for global alignment is based on the Needleman-Wunsch algorithm suitable for:
 - Sequences that are quite similar (more closely related)
 - Sequences of approximately same length

Local alignment - Focusing on just the best matching (highest scoring) regions

- The tool for local alignment is based on Smith-Waterman algorithm Suitable for:
 - Sequences similar along some of their lengths but dissimilar in others (i.e. sharing several conserved regions of local similarity/ domains)
 - Sequences that differ in length

A. Examples of Sequences

Sequence1 : ATACCGGATATT

Sequence2 : AACGGACCCT

B. Global Alignment

```
ATACCGGATATT
| | | | |
- - AACGGACCCT
      match
```

C. Local Alignment

```
ATACCGGATATT
| | | | |
- - AACGGA - - -
      match
```


De novo assembly

- Generally, there are two types of assembly methods: reference assembly and de novo assembly.
- The former one needs to use a reference genome, while de novo assembly does not.
- Through assembling the short continuous sequences called “contigs” by joining overlapping regions, de novo assembly is limited by the size and continuity of contigs

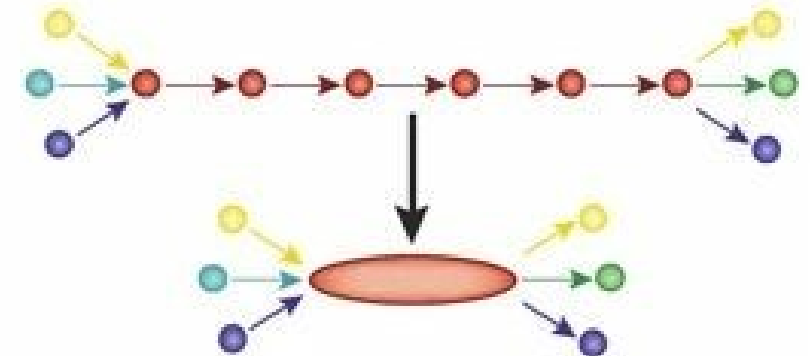
1. Fragment DNA and sequence



2. Find overlaps between reads

...AGCCTAGACCTACA**GGATGCGCGACACGT**
GGATGCGCGACACGTCGCATATCCGGT...

3. Assemble overlaps into contigs



4. Assemble contigs into scaffolds

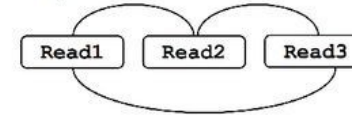


De novo assembly Algorithms

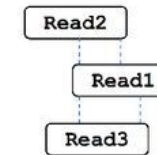
- The two most used algorithms for de novo assembly are overlap-layout-consensus (OLC) and de-bruijn-graph (DBG).
- OLC first finds the overlap regions, then generates the layout for all the reads and overlap information to make a graph to infer the consensus sequence.
- DBG cuts the reads into shorter overlapping k-mers, then uses these k-mers as vertices, connecting them through edges, to build a de Bruijn graph.
- By finding paths to visit each edge once, sequences would be assembled.
- In most cases, OLC is suitable for long reads such as Sanger Sequencing, while DBG performs better on short reads such as Illumina Sequencing

(a) Overlap, Layout, Consensus assembly

(i) Find overlaps



(ii) Layout reads



(iii) Build consensus

```

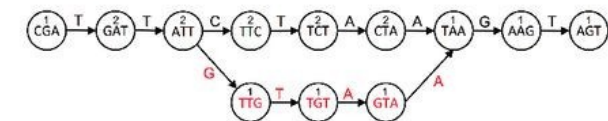
CGATTCTA
  TTCTAAGT
    GATTGTAA
  -----
CGATTCTAAGT
    
```

(b) De Bruijn graph assembly

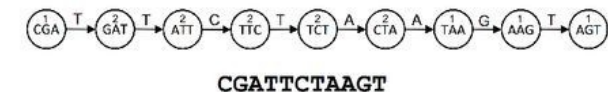
(i) Make kmers

Read1: TTCTAAGT	Read2: CGATTCTA	Read3: GATTGTAA
Kmers: TTC	Kmers: CGA	Kmers: GAT
TCT	GAT	ATT
CTA	AIT	TTG
TAA	TTC	TGT
AAG	TCT	GTA
AGT	CTA	TAA

(ii) Build graph



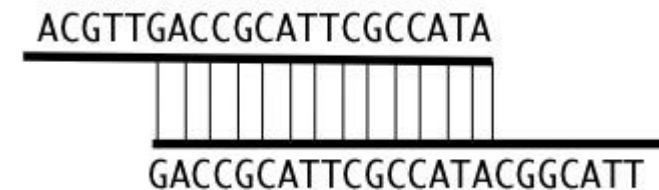
(iii) Walk graph and output contigs



Overlap layout consensus Algorithm

1. Overlap Detection (Overlap):

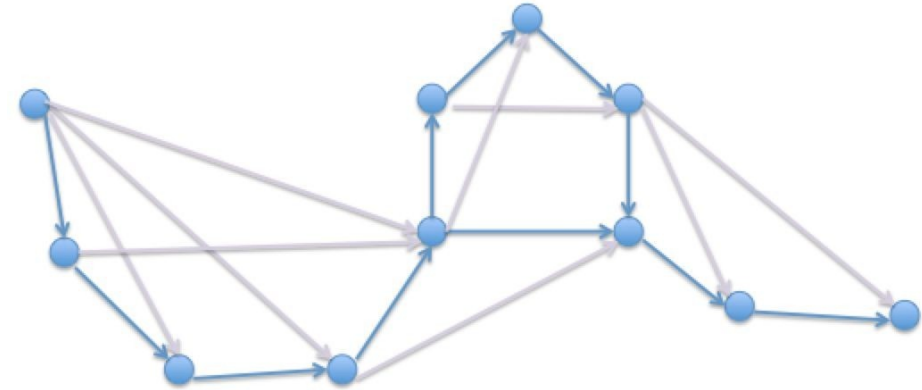
- The first step involves comparing all pairs of sequencing reads to identify regions of similarity or overlap between them. These overlaps are often referred to as "seed" overlaps.
- To combine the DNA fragments into larger segments, we must find places where two or more reads overlap, i.e. where the beginning sequence of one fragment matches the end sequence of another fragment. For example, given two fragments such as ACGTTGACCGCATTGCGCCATA and GACCGCATTGCGCCATACGGCATT, we can construct a larger sequence based on the overlap:



Overlap layout consensus Algorithm

2. Layout Generation (Layout):

- Once overlaps are detected, the next step is to construct a graph known as an overlap graph or a string graph.
- In this graph, each sequencing read corresponds to a node, and overlaps between reads are represented as edges connecting the nodes.
- The goal is to find a path or paths through this graph that connect all the reads.
- These paths represent a potential layout of the reads, indicating their order and how they overlap.
- There can be multiple paths through the graph, especially in complex regions with repeated sequences.
- Identifying the correct path or paths is a challenging aspect of the OLC approach.



Overlap layout consensus Algorithm

3. Consensus Sequence Generation (Consensus):

After obtaining a layout, the next step is to generate a consensus sequence for each path in the layout. This is done by aligning the reads along the chosen path and resolving discrepancies (e.g., sequencing errors) to create a single, high-quality consensus sequence.

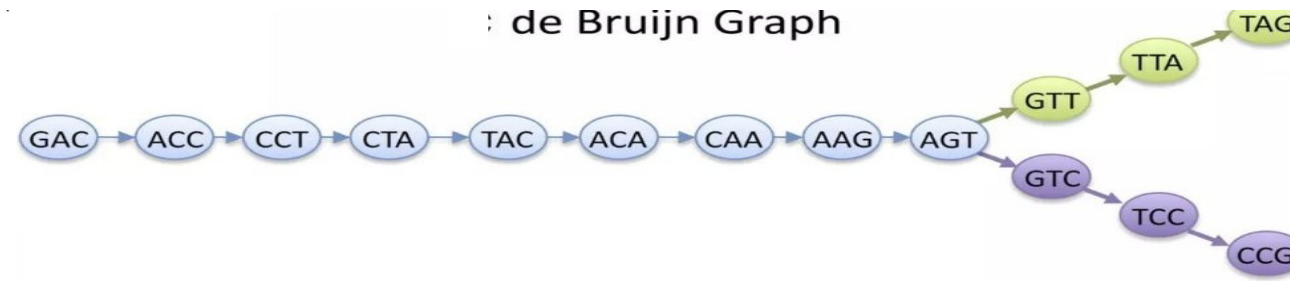
Various algorithms are used for consensus generation, and they often involve base-calling and quality score assessment at each position in the consensus.

4. Iterative Refinement (Optional):

Some OLC assemblers may perform multiple iterations of the process, improving the consensus sequences and the layout with each iteration. This iterative refinement can help resolve ambiguities and improve accuracy.

De bruijn graph Algorithm

- It breaks reads into successive k-mers and the graph maps the k-mers
- Each k-mer is a node and edges are drawn between each k-mer in a read
- Repeat sequences create a fork in the graph; alternative sequences create a bubble.
- The k-mer size can only be determined by “trial and error”.
- A small value of K will create a complex graph but a large value of K may miss small overlaps. A good starting point would be a k-mer size that is $\frac{2}{3}$ the size of the read
- Good for short reads or small genomes. With long reads and /or large genomes, may require lots of RAM (e.g. ~0.5 TB for human)
- Examples are: Velvet, SPades, SOAPdenovo, ALLPATHS-LG, ABySS

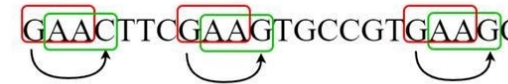


Key terminologies

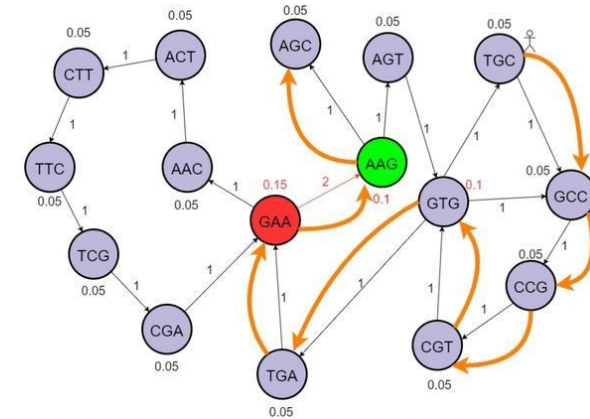
K-mers: In the context of DNA sequencing, a "k-mer" is a sequence of k consecutive nucleotide bases (A, C, G, or T). For example, if $k = 4$, you would have 4-mers like "ATCG," "GACT," and so on. The value of k is a parameter chosen by the user or the assembler and is typically referred to as the "k-mer size."

Building Blocks: De Bruijn graphs are constructed by considering all possible k-mers from a set of sequencing reads. These k-mers serve as the building blocks for the graph.

Nodes: Each k-mer becomes a node in the de Bruijn graph. Nodes represent the k-mers themselves, not the entire reads. For example, if you have the 4-mers "ATCG" and "TCGA," they would correspond to two nodes in the graph.



(a) A graph edge is directed from 3-mer node v_i to v_j if they overlap by two continuous nucleotides with v_i 's suffix being the v_j 's prefix, i.e., "GAA" (the red node in Fig. 2(b)) to "AAG" (the green node in Fig. 2(b)).



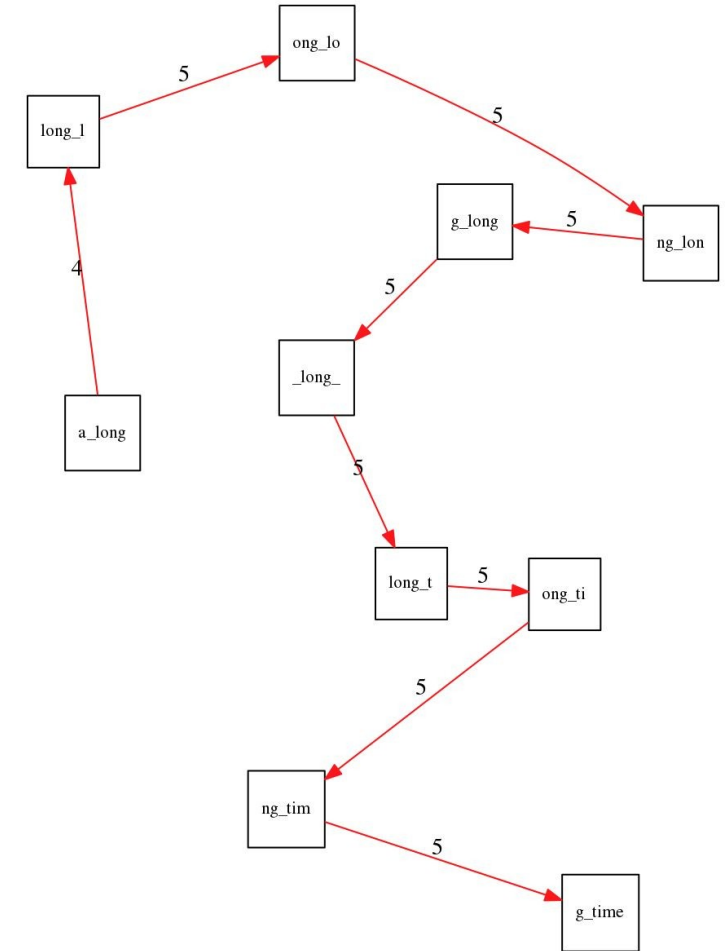
(b) The 3-mer graph built from the DNA sequence in Fig. 2(a). The solid line with arrows in yellow describes the path of a random walk with "TGC" as the starting node and "AGC" as the ending node. Noted that the pair of "GAA" to "AAG" occurs twice in the sequence, so that the weight of the graph edge from "GAA" to "AAG" is 2.

Key terminologies

Edges: Edges connect the nodes in the de Bruijn graph. Edges are formed when k-mers have an overlap of (k-1) bases. For example, if "ATCG" and "TCGA" overlap by three bases, they would be connected by an edge.

Paths: De Bruijn graphs contain paths through the nodes, where each path represents a potential sequence. By traversing the graph and selecting an appropriate path, you can reconstruct a sequence.

Assembly: The main goal of constructing a de Bruijn graph is to find a path through the graph that covers all nodes (k-mers), thereby reconstructing a full sequence (e.g., a genome or a contig). This path corresponds to the assembled sequence.



Choosing the Approach

Reference assembly:

- When reference genome exists and is reliable.
- Example: clinical diagnostics (HIV, SARS-CoV-2).

De novo assembly:

- For novel pathogens, outbreak investigations, metagenomics.
- Example: discovery of new Klebsiella plasmids in wastewater.
- Hybrid approaches possible (reference-guided de novo).

Quality assessment of Assemblies

- There is no individual metric that exists to determine if we have a good assembly or not, especially if we have no reference.
- There are some general things we can look at, like N50 or largest contig, or fraction of reads that successfully recruit to our assembly, and more.
- But for one, these don't really have any solid context to know if they're "good" or not unless we're comparing multiple assemblies of the same data.
- So we have to keep in mind what our goals are, and know that picking the "best" assembly is not a trivial or straightforward task.
- Having a reference genome like we do in this case however, can make things a lot easier.

Quality assessment of Assemblies

1. N50 / L50 (assembly contiguity):

- Given a set of contigs, the **N50** is defined as the sequence length of the shortest contig at 50% of the total assembly length.
- Given a set of contigs, each with its own length, the **L50** is defined as count of smallest number of contigs whose length sum makes up half of genome size.

2. Total length vs expected genome size.

3. Number of contigs/scaffolds (fewer = better).

4. BUSCO scores (completeness of conserved genes).

5. GC content consistency.

Quality assessment of Assemblies

- A way to calculate metrics assembly is to use **QUAST = Quality ASsessment Tool**. Quast is a tool to evaluate genome assemblies by computing various metrics and to compare genome assembly with a reference genome.

QUAST will provide us with the following statistics:

- **Contigs:** The total number of contigs in the assembly.
- **Largest contig:** The length of the largest contig in the assembly.
- **Total length:** The total number of bases in the assembly.
- **Nx:** The largest contig length, L, such that contigs of length $\geq L$ account for at least x% of the bases of the assembly.
- **NGx:** The contig length such that using equal or longer length contigs produces x% of the length of the reference genome, rather than x% of the assembly length.
- **GC content:** the percentage of nitrogenous bases which are either guanine or cytosine.

Quality assessment of Assemblies

If a reference genome is provided, the report also indicates:

- **Genome fraction (%)**: total number of aligned bases in the reference genome, divided by the genome size
- **Duplication ratio**: total number of aligned bases in the assembly, divided by the total number of aligned bases in the reference genome
- **NGA50**: NG50 where the lengths of aligned blocks are counted instead of contig lengths. i.e., if a contig has a misassembly with respect to the reference, the contig is broken into smaller pieces. This metric is computed only if a reference genome is provided.
- **LGA50**: LG50 where aligned blocks are counted instead of contigs. i.e., if a contig has a misassembly with respect to the reference, the contig is broken into smaller pieces

Evaluation of assembly completeness

Core genes completeness with BUSCO

- **BUSCO (Benchmarking Universal Single-Copy Orthologs)** allows a measure for quantitative assessment of genome assembly based on evolutionarily informed expectations of gene content.
- Despite BUSCO being robust for species that have been widely studied, it can be inaccurate when the newly assembled genome belongs to a taxonomic group that is not well represented in OrthoDB.
- Even in a well-represented taxonomic group, the bias on the selection of reference genomes selected to create OrthoDB can lead to an under-scoring of the newly assembled genome and is dependent on the evolution of the genomes.
- For example, in microsporidia, basal genomes have much lower scores due to the strong drive on gene loss and gain in these organisms.